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Project

SW321: Software Architecture and Design

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Topic: Smart Home System Architecture

O61 Section

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Q1.Functional Requirements

A. Security Requirements

1. The system shall allow homeowners to log in using a secure username and password before accessing the smart home dashboard.
2. The system shall authorize users based on their roles (homeowner or guest) to control specific devices.

B. Monitoring and Alert Requirements

1. The system shall monitor motion sensors and send an intrusion alert to the homeowner within 5 seconds of detection.
2. The system shall detect smoke or gas leaks and immediately notify the homeowner through a mobile notification.

C. Control Requirements

1. The system shall allow users to remotely lock and unlock smart doors using the mobile app.
2. The system shall allow users to turn smart lights on or off and change the temperature control remotely.

D. Exception and Alternative Case Requirements

1. The system shall notify the user when a smart device goes offline or becomes unreachable.
2. The system shall continue working without internet and send saved alerts after the connection is restored.

Q2.Quality attributes scenarios:

A. Performance Scenario (Motion Sensor):

- Stimulus source: Motion sensor.
- Stimulus: Detects movement in the hallway.
- Environment: Normal operation.
- Artifact: Smart lights.
- Response: The system processes the signal and triggers the lighting control.
- Response measure: The smart lights must turn on within 200 milliseconds of the motion detection.

B. Availability Scenario (Hub Power Failure):

- Stimulus source: power failure.
- Stimulus: The main power supply to the Smart Home Hub is disconnected.
- Environment: Normal operation.
- Artifact: Smart Home Hub.
- Response: The system switches to an internal battery backup and sends a "Power Loss" alert to the mobile app.
- Response measure: The system must remain less than 10% of failure rate on backup power for at least 3 hours.

C. Security Scenario (Unauthorized Access):

- Stimulus source: External unauthorized user.
- Stimulus: Attempts to log into the Mobile App using incorrect credentials.
- Environment: Remote access via the internet.
- Artifact: Authentication Service.
- Response: The system logs the failed attempt and temporarily disables the account.
- Response measure: The account must be locked for 10 minutes after 3 consecutive failed attempts.

D. Modifiability Scenario (UI Change):

- Stimulus source: System Developer.
- Stimulus: A request to change the Mobile Dashboard theme from "Light Mode" to "Dark Mode".
- Environment: Design and development time.
- Artifact: Mobile App / Web Dashboard.
- Response: The developer modifies the UI styling layer without changing core logic.
- Response measure: The change must be completed within 3 hours of development time.

Q3.Architectural Style Selection and Justification:

For the smart home system, the Sense–Compute–Control architectural style has been selected due to its suitability for event-driven and real-time control applications.

Why this style fits Smart Home

The Sense–Compute–Control architectural style is well suited for a smart home system because it follows a clear sequence of sensing environmental changes, making decisions, and then executing actions. Smart home systems depend on sensors to detect events such as motion, smoke, gas leakage, or temperature changes. These sensed events are processed by the system to determine the appropriate response, such as turning on lights, locking doors, or sending alerts. Therefore, the Sense–Compute–Control style naturally matches the operational needs of smart home systems.

Support for Functional Requirements (Q1)

The Sense–Compute–Control architectural style directly supports the functional requirements of the smart home system.

The sensing component supports monitoring and alert requirements by continuously collecting data from motion sensors, smoke detectors, and other smart devices.

The compute component analyzes the sensed data, applies system rules and security policies such as authorization, and determines the appropriate system response.

The control component fulfills control requirements by executing actions such as controlling lighting, adjusting temperature, locking or unlocking doors, and sending notifications.

In addition, exception and alternative cases, such as device disconnection or internet failure, can be handled by the compute component through alternative actions or stored alerts.

Support for Quality Attributes (Q2)

The Sense–Compute–Control architecture also supports the defined quality attribute scenarios.

Performance is enhanced by processing sensor data locally and making fast decisions, enabling quick responses such as turning on lights within milliseconds.

Availability is supported by allowing the system to continue essential operations during power or network failures through local control mechanisms and backup solutions.

Security is strengthened by centralizing authentication, authorization, and logging within the compute component.

Modifiability is improved because sensors and control devices can be modified or extended with minimal impact on other system components as long as interfaces remain consistent.

Tradeoffs of the Selected Architectural Style

Improves: The Sense–Compute–Control style improves system responsiveness and clarity by clearly separating sensing, decision-making, and execution, which is essential for real-time smart home operations.

Harder: As the system scales and becomes more complex, the compute component may require additional effort to manage and optimize in order to avoid performance bottlenecks.

Q4.Component & Connector (C&C)

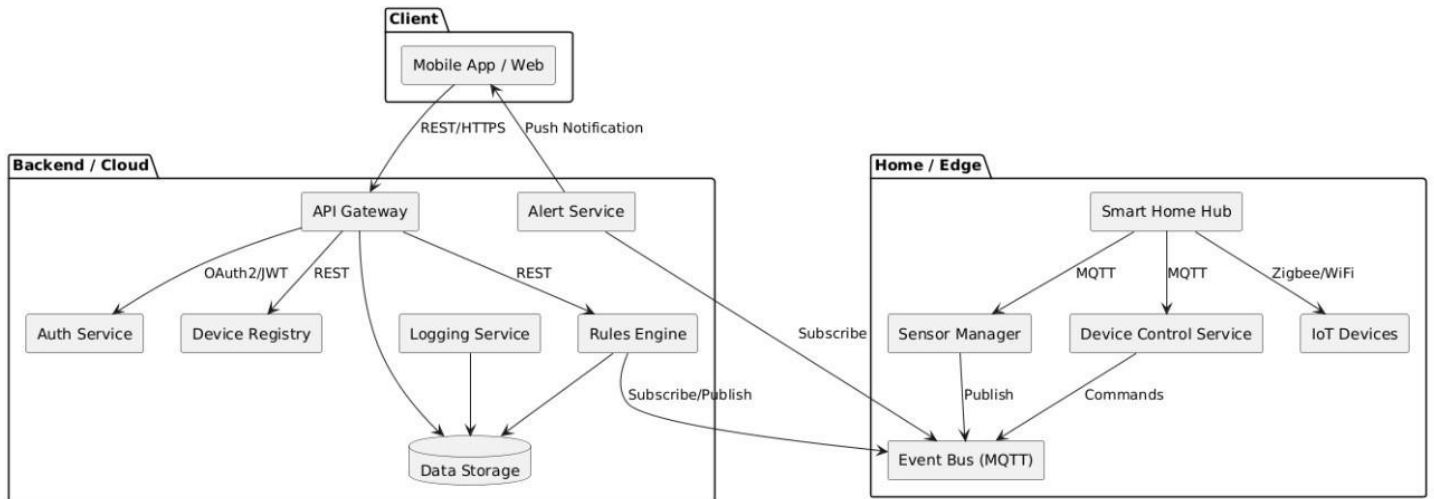


Figure 1: The system is modeled using a Component-and-Connector (C&C) view.

It includes backend, client, and home components that communicate using REST and MQTT connectors.

Q5.Design Patterns

5.1 Scenario 1:

Scenario A:

Design pattern: Adapter pattern.

Why it fits: The Adapter pattern allows the system to interact with various vendor-specific APIs through a single, unified interface, making it possible to add new device types without modifying existing core logic.

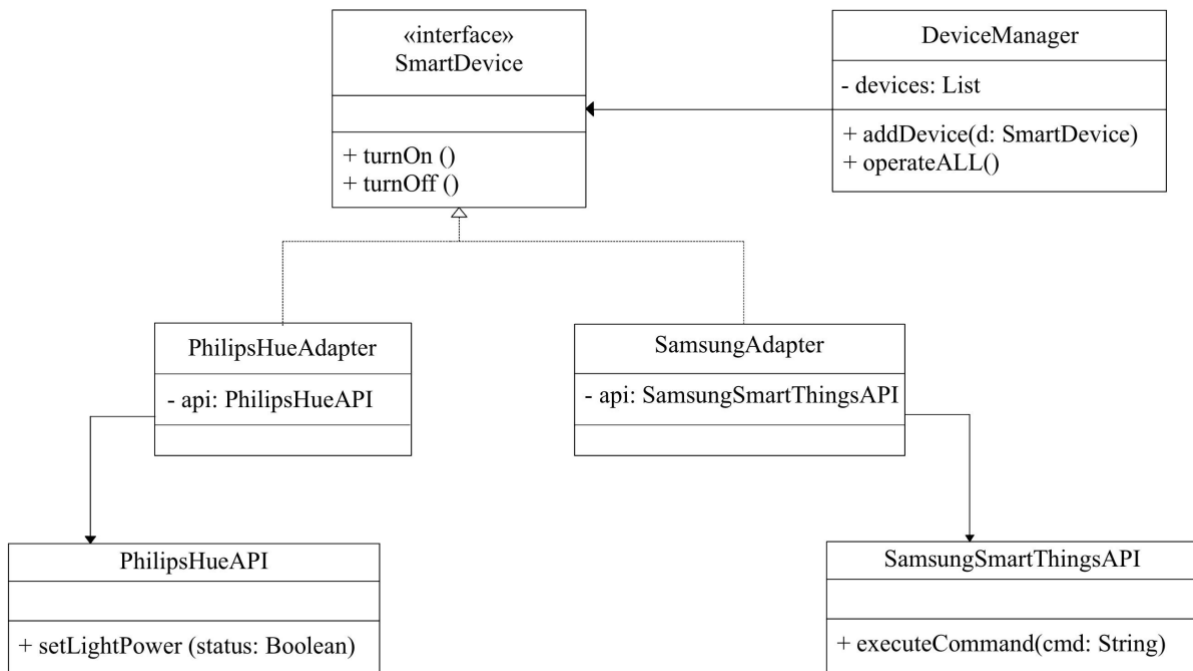


Figure 2

5.2 Scenario 2:

Scenario B:

Design pattern: Observer pattern.

Why it fits: The observer pattern establishes a one-to-many relationship that ensures all dependent services are automatically notified and updated the moment a device's status changes.

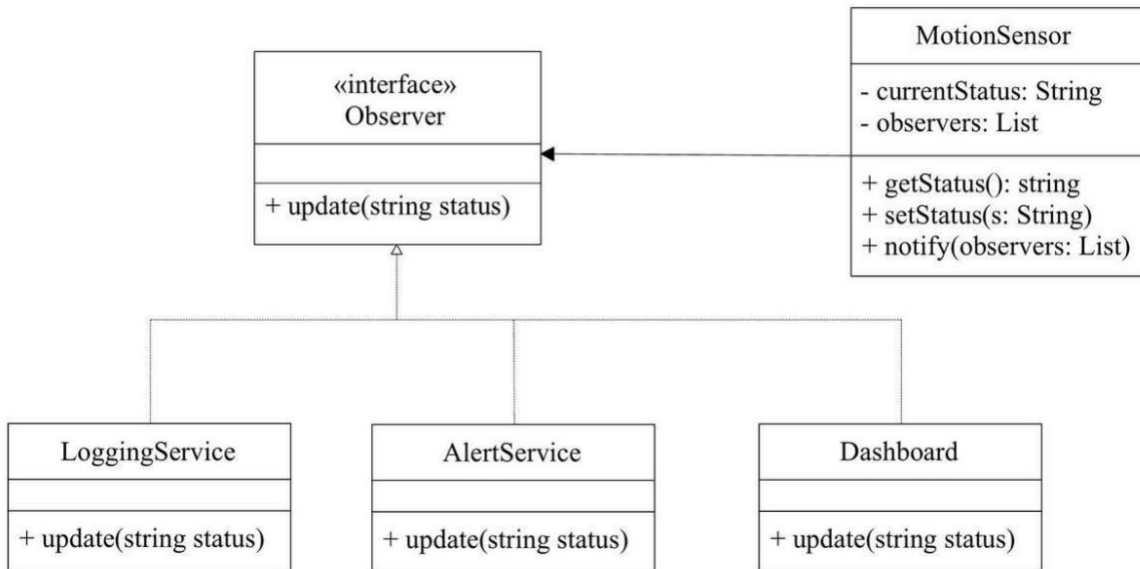


Figure 3

Q6.UML Documentation

6.1-Use case diagram

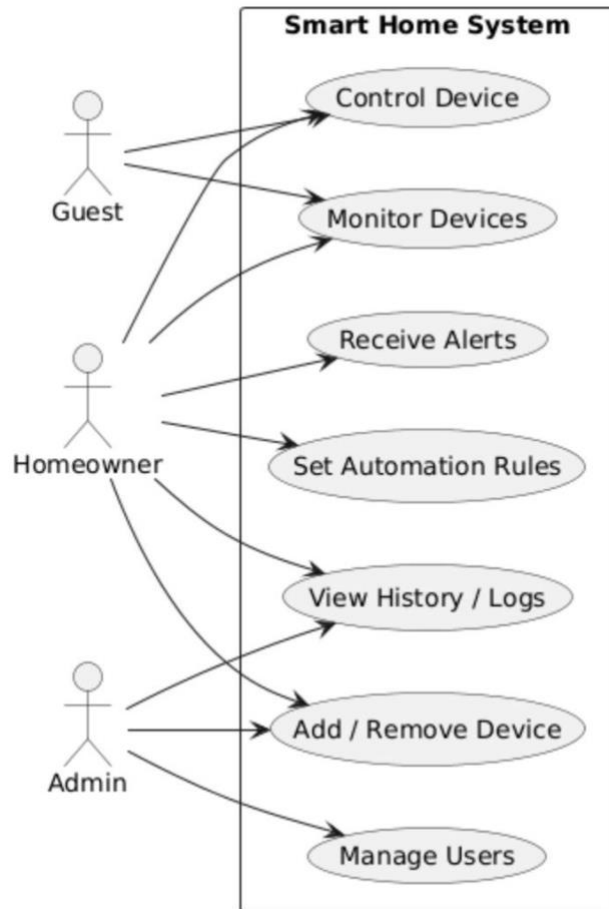


Figure 4: The following Use Case Diagram illustrates the interactions between users and the Smart Home System.

6.2-Component diagram

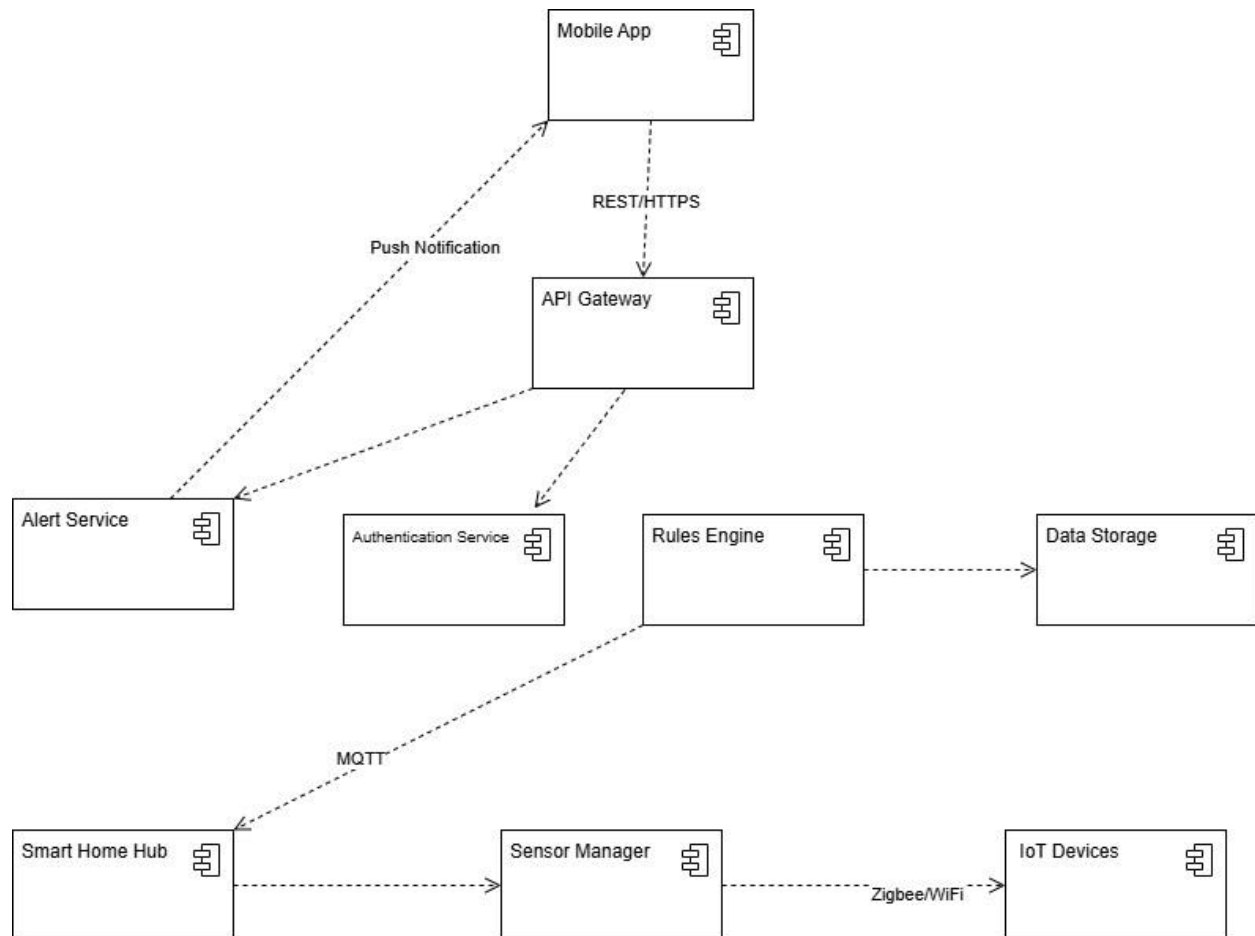


Figure 5: This diagram shows the main components of the Smart Home System and how they depend on each other. The Mobile App communicates with backend services through the API Gateway. The backend connects to home devices via the Smart Home Hub to manage and monitor IoT devices.

6.3-Deployment diagram

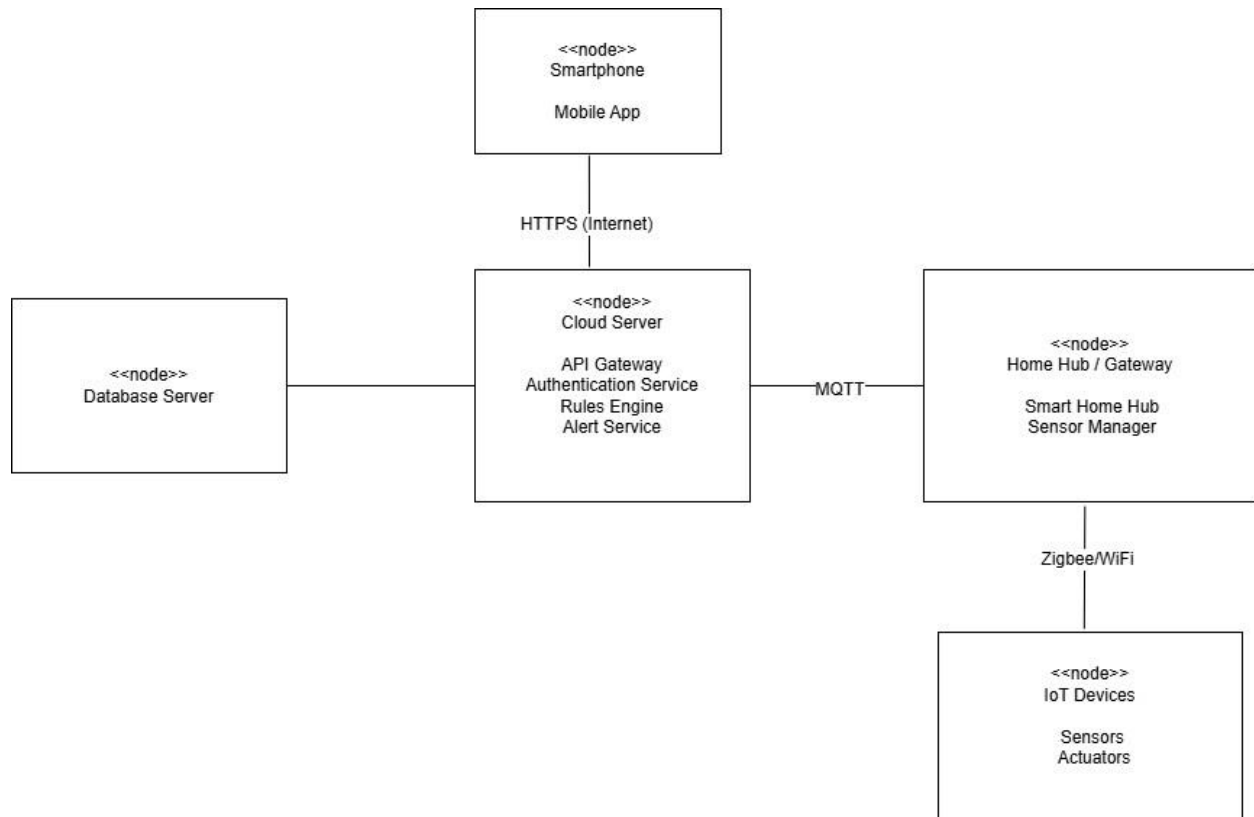


Figure 6: This deployment diagram shows the physical architecture of the Smart Home system. The Mobile App communicates with the Cloud Server via HTTPS, which hosts the backend services and connects to the Database Server. The Cloud Server interacts with the Home Hub using MQTT, and the Hub communicates with IoT devices via Zigbee/Wi-Fi.

7. Evaluation Preparation: Drivers + Architectural Approaches

7.1 Architectural Drivers

1. Security
The system must protect smart home access by using secure login, role-based authorization, and protection against unauthorized access attempts.
2. Performance
The system must respond quickly to sensor events, such as turning on smart lights within 200 milliseconds after motion detection.
3. Availability
The system must continue essential operations during failures, such as power loss or internet disconnection, by using backup power and local control.
4. Modifiability
The system should allow new smart devices, vendors, or UI changes to be added with minimal changes to the core system logic.
5. Monitoring and Alerting
The system must continuously monitor smart devices and send alerts to the homeowner when motion, smoke, gas leaks, or device failures are detected.

7.2 Architectural Approaches / Decisions

1. Sense–Compute–Control Architectural Style
The system uses this style because smart home systems depend on sensing events, processing them, and then controlling devices or sending alerts.
2. Event-Driven Communication using MQTT
MQTT is used between the Smart Home Hub, sensors, devices, and backend services to support fast event publishing and subscription.
3. Authentication and Authorization Service
An Auth Service is used to verify user identity and control access based on user roles such as homeowner, guest, and admin.
4. Logging and Monitoring Service
The system includes a Logging Service to record device activities, failed login attempts, alerts, and system events for security and maintenance.
5. Adapter Pattern for Device Vendors
The Adapter pattern allows the system to support different smart device vendors through one common interface without changing the core logic.
6. Observer Pattern for Device Status Updates
The Observer pattern is used so that services such as Alert Service, Logging Service, and Dashboard can be automatically updated when a device status changes.

Q8.Utility Tree: Prioritized Quality Scenarios

Quality Attribute	Scenario ID	Quality Scenario	Measurable Response	Priority
Performance	P1	When a motion sensor detects movement while the security mode is active, the system shall process the event and send an intrusion alert to the homeowner.	The alert must be delivered to the homeowner's mobile app within 5 seconds of motion detection.	H
Performance	P2	When the homeowner opens the mobile dashboard to monitor devices, the system shall retrieve and display the current status of all connected smart devices.	The dashboard must display device statuses within 2 seconds for up to 20 connected devices.	M
Availability	A1	When the internet connection is lost, the Smart Home Hub shall continue operating essential local functions such as door lock control, smoke detection, and motion-based automation.	Essential local functions must remain available for at least 3 hours during internet outage.	H
Availability	A2	When the cloud service becomes unavailable, the system shall store sensor events and alerts locally until the cloud connection is restored.	The system must store at least 500 events locally and synchronize them within 2 minutes after the cloud service is restored	M
Security	S1	When an unauthorized user attempts to log in to the mobile app using incorrect credentials, the Authentication Service shall detect and restrict repeated failed attempts.	The account must be locked for 10 minutes after 3 consecutive failed login attempts.	H
Security	S2	When a guest user tries to control restricted devices such as smart door locks or security cameras, the Authorization Service shall deny the action and record the attempt.	Unauthorized actions must be denied within 1 second, and the attempt must be logged within 2 seconds.	H

Modifiability	M1	When a new smart device vendor is added, the system shall support the new vendor through the Adapter Pattern without changing existing device-handling logic.	The new vendor integration must require changes to no more than 2 components and be completed within 1 working day.	H
Modifiability	M2	When the development team needs to change the mobile dashboard theme or layout, the system shall allow the change without modifying backend services or device-control logic.	The UI change must be completed within 3 hours by modifying only the UI styling layer.	L

Table 1: Utility Tree for the Smart Home System. The table presents the main quality attributes of the system, including Performance, Availability, Security, and Modifiability. Each quality attribute includes two measurable scenarios with a priority level of High (H), Medium (M), or Low (L) based on its importance to the system.

Q9.Scenario Walkthrough + Result (Risks & Themes)

Scenario 1: Intrusion Alert Response (P1)

- Scenario Description: A motion sensor detects movement while security mode is active, and the system must send an intrusion alert to the homeowner's mobile app within 5 seconds.
- Architectural Decisions Involved:
 - Style: Sense-Compute-Control.
 - Connectors: MQTT Event Bus, Push Notification (REST/HTTPS).
 - Components: Sensor Manager, Smart Home Hub, Rules Engine, Alert Service.
- Satisfied? Partial, while the Sense-Compute-Control style allows for fast local detection and rule processing at the Hub, the delivery of the final alert depends on the Cloud-based Alert Service and external internet connectivity. If the homeowner's internet or the cloud service is down, the 5-second notification target cannot be guaranteed.
- Sensitivity Point: The response time for safety-critical alerts is highly sensitive to the availability and latency of the cloud-to-mobile communication path.

Scenario 2: Hub Power Failure (A1)

- Scenario Description: The main power to the Smart Home Hub is disconnected. The system must switch to battery backup and remain operational for at least 3 hours.
- Architectural Decisions Involved:
 - Tactics: Active Redundancy (Battery), Fault Masking.
 - Components: Smart Home Hub, Alert Service.

- Satisfied? Partial, While the Smart Home Hub includes an internal battery, the deployment diagram shows the Alert Service and Rules Engine residing in the Cloud Server. If a power failure also disables the home's internet router, the Hub can maintain local control, but the homeowner may not receive the "Power Loss" mobile notification until the connection is restored.
- Tradeoff: Using a local Hub increases availability during power outages, but relying on Cloud-based Alerting creates a dependency on internet connectivity for remote notifications.

Scenario 3: Unauthorized Access (S1)

- Scenario Description: An external unauthorized user attempts to log into the Mobile App using incorrect credentials three times. The system must lock the account for 10 minutes.
- Architectural Decisions Involved:
 - Components: API Gateway, Authentication Service, Logging Service.
 - Tactics: Authenticate Actors, Manage User Model, Record/Log.
- Satisfied? Yes, the architecture centralizes security within the Backend/Cloud layer. The API Gateway routes requests to the Authentication Service, which manages session logic and lockout states, while the Logging Service maintains a record of failed attempts to trigger the lock.
- Non-risk: Centralizing authentication and authorization ensures that security policies are applied consistently across all access points (Mobile and Web).

Summery:

Risk ID	Risk Description	Risk Theme
R1	Dependency on cloud-based Rules Engine for alerts may prevent critical notifications during internet outages.	Reliability & Connectivity
R2	The Smart Home Hub is a single point of failure for all local Zigbee/WiFi device communications.	Reliability & Connectivity
R3	High volume of sensor data could create a performance bottleneck in the MQTT Event Bus.	Scalability

Risk Themes & Mitigations

- Theme 1: Reliability & Connectivity
 - Mitigation: Implement a "Local-First" logic approach where the Rules Engine is mirrored on the Smart Home Hub. Additionally, add a cellular failover module (GSM) to the Hub to ensure alerts are sent even if the primary internet/power fails.
- Theme 2: Scalability
 - Mitigation: Use a distributed MQTT broker architecture or implement "Edge Filtering" in the Sensor Manager to discard irrelevant data before it reaches the Event Bus, reducing the processing load on the Hub.

